

Multiple Choice

1. **Answer:** A

Options: A) 22atm; B) 20atm; C) 15atm; D) 30atm; E) 34atm

Note: Correct option: A - 22atm

2. **Answer:** B

Options: A) 0.9; B) 1.0; C) 2.5; D) 0.3; E) 0.5

Note: Correct option: B - 1.0

3. **Answer:** A

Options: A) 2×10^{-10} m; B) 1×10^{-10} m; C) 7×10^{-10} m; D) 6×10^{-10} m; E) 4×10^{-10} m

Note: Correct option: A - 2×10^{-10} m

4. **Answer:** A

Options: A) 169 gmol^{-1} ; B) 123 gmol^{-1} ; C) 200 gmol^{-1} ; D) 150 gmol^{-1} ; E) 190 gmol^{-1}

Note: Correct option: A - 169 gmol^{-1}

5. **Answer:** C

Options: A) lead; B) cesium; C) bromine; D) francium; E) rubidium

Note: Correct option: C - bromine

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '6500 cubic feet'.

Long Form Response

11. **Answer:** 45.0 ml. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Density of the object is 3.33 g/ml and the oil is 0.83 g/ml. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.